'5bua suita n east to so@ooo 00 0 o0 o::. ~~Nnp~~

' O'

$$\underbrace{\begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 3 & 4 \end{bmatrix}}_{3 \times 2} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{\mathbb{R}^2} = \underbrace{\begin{bmatrix} 0 \\ 1 \\ -6 \end{bmatrix}}_{\mathbb{R}^3}$$

$$x_1 + x_2 + x_3 = 1 \qquad \underbrace{\begin{bmatrix} 1 & 1 & 1 \end{bmatrix}}_{1 \times 3} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{\mathbb{R}^3} = \underbrace{1}_{\mathbb{R}^1}$$

$$A\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

when does solution exist.

a transformation from $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ via A to $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

$$\boxed{\begin{aligned} a_{11}x_1 + a_{12}x_2 &= b_1 \\ a_{21}x_1 + a_{22}x_2 &= b_2 \\ a_{31}x_1 + a_{32}x_2 &= b_3 \end{aligned}}$$

does $\exists$ a $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ which ~~trans~~ under A results in b

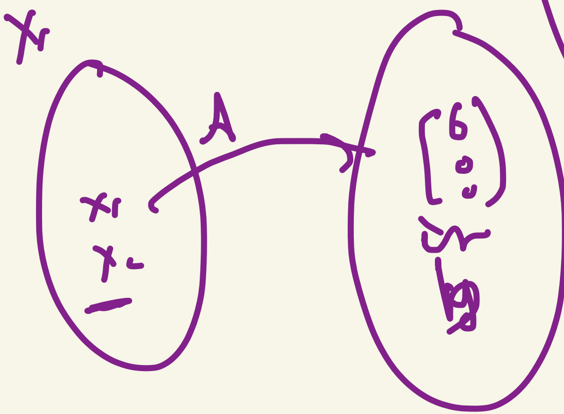
$$U \quad ?x \xrightarrow{A} b \quad V$$

$$\underbrace{\begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 2 & 0 \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}}_{x} = \underbrace{\begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix}}_{b}$$

U

there is no x s.t $Ax = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix}$

$\begin{bmatrix} 6 \end{bmatrix}$



b must be in the Im.(A)

there is NO x such that

b is not in Im(A)

① $\underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}}_{A} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \underbrace{\begin{bmatrix} 2 \\ 4 \end{bmatrix}}_{b}$

$$\boxed{\begin{aligned} x_1 &= 2 \\ x_2 &= 2. \end{aligned}}$$

is the solution: unique!

$\underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$

$x_1 = 4$

no solution can

$0 \cdot x_2 = 2$

$\begin{pmatrix} 4 \\ 2 \end{pmatrix}$ does not lie in Im $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

③ $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$

$x_1 = 0$ & x_2 can be anything

ker $\begin{pmatrix} 0 \\ * \end{pmatrix}$

③ $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$

$x_1 = 4$

$0 \cdot x_2 = 0$

$(4, 0)$, $(4, 4)$, $(4, -6)$ - -

Case 3 has inf. solution

$A x = 0$

$x = 0$

Ax is always zero

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

is $x = 0$ the ONLY solution:

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$x_1 = x_2 = 0$

is this ONLY solution

$f : U \to V.$

$\mathbb{R}^n \qquad \mathbb{R}^m$

$U = \{u_1 \;.\; u_n\}$

$$\begin{pmatrix} c_{11} & c_{21} \\ c_{21} & c_{22} \\ | & | \\ c_{m1} & c_{m-} \end{pmatrix}$$

Span / basis

$f(u_k)$ will be a vector in V.

$$f(u_1) = c_{11} v_1 + c_{21} v_2 + \cdot \; c_{m1} v_m$$

$$f(u_k) = \sum_{i=1}^{m} c_{ik} v_i$$

how does any vector in V look like

where $c_{1k}, \ldots, c_{mk}$ are the coordinates of $f(u_k)$ wrt basis $(v_1 \ldots v_m)$

any vector in V can be written as a linear comb. of $v_1 \ldots v_m$

$f: \underset{x}{U} \to \underset{y}{V}$: the corresponding matrix $A_{m\times n}$.

$\underline{\dim \operatorname{Ker}(f)} + \dim \operatorname{Im}(f) = n$

$\operatorname{rank}(A) + \operatorname{null}(A) = n$

$\operatorname{Ker} f \subset \mathbb{R}^n$

$\boxed{\dim \operatorname{Ker} f : k < n.}$ (basis $\{e_1 \ldots e_k\}$ ← basis for Ker f

Since U is $\mathbb{R}^n$ $\{e_1 \ldots e_k, e_{k+1} \ldots e_n\}$ ← basis for $\mathbb{R}^n$

basis for Im f

$\boxed{f(e_i),\ i = 1, \ldots k,\ = 0}$

$\boxed{f(e_{k+1}) \ \ldots \ f(e_n)}$ ← span Im f.

$y = f(x) = f(\cancel{c_1 e_1} + \ldots + \cancel{c_k e_k} + c_{k+1}e_{k+1} + \ldots c_n e_n)$

$x \in \mathbb{R}^n,\ y \in \mathbb{R}^m$ $\quad \underline{c_{k+1} f(e_{k+1}) + \ldots + c_n f(e_n)}$

$$\boxed{c_{k+1} f(e_{k+1}) + \dots + c_n f(e_n) = 0}$$ iff $c_{k+1} \dots c_n = 0$

linear combination of vectors $f(e_{k+1}) \dots f(e_n)$

$$\underbrace{f(c_{k+1} e_{k+1} + \dots + c_n e_n)}_{x} = 0.$$

A

is in the ker f.

$\underbrace{(e_{k+1} \dots e_n)} \Leftarrow\Rightarrow \underbrace{(e_1 \dots e_n)}$

CONTRADICTION. $c_{k+1} \dots c_n \neq 0$

$$c_{k+1} e_{k+1} + \dots + c_n e_n \neq 0$$

$e_{k+1} \dots e_n$ are linearly independent

Im f. $(k) + (n-k) = n.$

$$\boxed{Ax = 0}$$

$x \neq 0$

$\mathbb{R}^n$

$\{e_1 \dots e_k, e_{k+1}, \dots e_n\}$

linearly IND

$f : U \to V$

$\mathbb{R}^n \quad \mathbb{R}^m$

$\dim \operatorname{Im}(f) + \underline{\dim \ker(f)} = n$

$\operatorname{Rank}(A) + \operatorname{null} A = n$

$e_1 \ldots e_n$

$\dim \ker f = k$

$(e_1 \ldots e_k, e_{k+1} \ldots e_n)$

$(n-k) + k = n$

$f(e_1)$

$\vdots$

$f(e_k)$

$= 0$

$\underbrace{f(e_{k+1}) \ldots f(e_n)}$ $\neq 0$

linearly independent

basis for the Im f

K

$n - k$

Rank Nullity Theorem

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 2 \\ 2 & 3 \end{bmatrix}$$

how many Ind C ②

$$\underbrace{\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 2 & 3 \end{bmatrix}}_{A^T}$$ ②

dim $C(A)$

dim $R(A)$

$= n.$

$$\begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

rank A + null A = 2

rank A^T + null A^T = 4

$$\boxed{\text{rank}(A) + \text{null}(A) = n.}$$

$$\underset{②}{2} \quad \underset{②}{0} \quad = \underset{④}{2}$$

rank A. ②

$C(A) = 1$

$$\begin{bmatrix} 1 & 2 & 0 & 0 \\ 1 & 2 & 0 & 0 \end{bmatrix}$$

$R(A) = 1$

A

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 2 & 3 \end{bmatrix}$$

what is the null A

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0$$

A

Ker.

Image

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{null } A$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \quad \leftarrow \text{right}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \in N(A)$$

$$\left(\begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \right)^T = 0$$

Left null space of A

$$y^T A = 0 \Rightarrow A^T y = 0$$

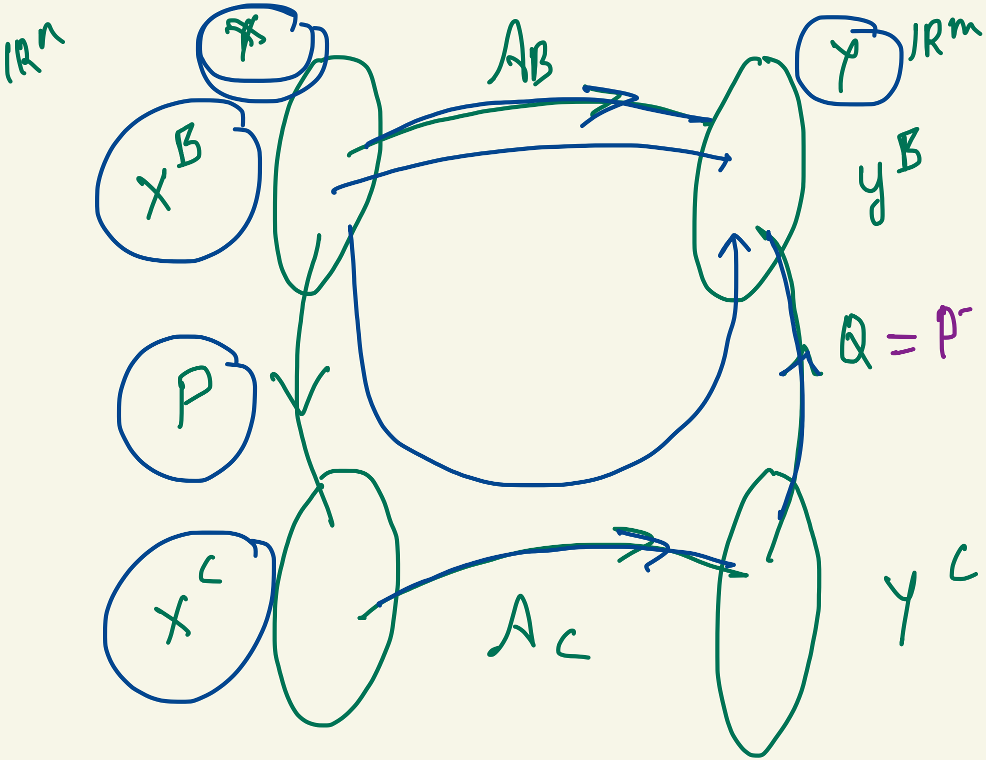
$$y \in \mathrm{Ker}(A^T)$$

= null space of A^T

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{A^T} \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}}_{y} = 0$$

$y \in \mathrm{Ker}$

null (A^T)



$A_B \cdot x^B = y^B.$

$A_B = P \cdot A_C \cdot Q.$

$A_B = P \cdot A_C P^{-1}$

$A_C = P^{-1} A_B P$

$A_C : \mathbb{R}^n \to \mathbb{R}^n$

$A_B : \mathbb{R}^n \to \mathbb{R}^n$

Special case

$X = Y \qquad \mathbb{R}^n \to \mathbb{R}^n$

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix} = 2\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3\begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$$= \frac{5}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{1}{2}\begin{bmatrix} -1 \\ 1 \end{bmatrix} \rightarrow \begin{pmatrix} 5/2 \\ 1/2 \end{pmatrix}$$

$$\underbrace{y^T A}_{} = 0 \rightarrow 0 = \begin{bmatrix} 0 \\ \vdots \end{bmatrix}$$

$\rightarrow$ 0 vector

$$\underbrace{A^T y}_{} = 0$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} =$$

$$\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}\Big[$$

$$x + 2y + z = 2$$
$$3x + 8y + z = 12$$
$$4y + z = 2$$

$$\left[\begin{array}{ccc|c} \boxed{1} & 2 & 1 & 2 \\ 3 & 8 & 1 & 12 \\ \boxed{0} & 4 & 1 & 2 \end{array}\right]$$

$$\begin{bmatrix} * & * & * \\ 0 & * & * \\ 0 & 0 & -* \end{bmatrix}$$

LU Decompo sition

QR decomp.

$R_2 \to R_2 - 3R_1$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 6 \\ 0 & \boxed{4} & 1 & 2 \end{array}\right]$$

$R_3 : R_3 - 2R_2$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 6 \\ 0 & 0 & 5 & -10 \end{array}\right]$$

$$x + 2y + z = 2$$
$$2y - 2z = 6$$
$$\boxed{5z = -10}$$

$z = -2$; $y = 1$.

$x = 2$

A: λ_1, λ_2.
e_1, e_2.

Column

$A e_1 \quad A e_2$

Colun

$A \quad \lambda_1 \ldots \lambda_n$
$e_1 \ldots e_n$

$$A e_1 = \lambda_1 e_1$$
$$A e_2 = \lambda_2 e_2.$$

$A [e_1 \; e_2]$
$= \lambda_1 e_1 \; \lambda_2 e_2$

$$\underline{A} [e_1 \; e_2] = \boxed{[e_1 \; e_2] \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}}$$

$$A [e_1 \ldots e_n] = [e_1 \ldots e_n] \begin{bmatrix} \lambda_1 & 0 & \cdot \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{bmatrix}$$

$$A E = E D$$

D is a diagonal matrix with eigen value of A ($\lambda_1 \ldots \lambda_n$) on its diagonal.

$$A e_1 = \lambda_1 e_1$$
$$\begin{bmatrix} \underset{c_1}{A e_1} & \underset{c_2}{A e_2} \end{bmatrix}$$

$$[a \quad b] = [c \quad d]$$

$$A e_1 = \lambda_1 e_1$$

$$A e_2 = \lambda_2 e_2$$

$$[A e_1 \quad A e_2] = \lambda_1 e_1 \quad \lambda_2 e_2$$

$$A \underbrace{[e_1 \quad e_2]} = (e_1 \quad e_2)\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

$$A E = E D \qquad E = [e_1 \quad e_2 \dots e_n]$$

$$A = E D E'$$

A^n.

$$\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}^n$$

$$\begin{pmatrix} \lambda_1^n & 0 \\ 0 & \lambda_2^n \end{pmatrix}$$

$$A = E D E^{-1}$$

$$A^2 = (E D E^{-1})(E D E^{-1})$$

$$= E D^2 E^{-1}$$

instead of n matrix mult

only comput. n^{th} power of $\lambda_1 \cdot \lambda_2$ & Two matrix Multiplication

$$A^n = \underline{E D^n E^{-1}}$$

computationally cheaper.

$$A = E \cdot D \cdot E^{-1}$$

$$A^2 = \underbrace{E \cdot D \cdot E^{-1} \cdot E}_{} \cdot D \cdot E^{-1}$$

$$E \quad D \quad I \quad D \quad E^{-1}$$

$$= E D^2 E^{-1}$$

$A \cdot A^T = A^2$

iff

A is

Symmetric

$A \cdot A \cdot A$

$A^2 = A \cdot A$

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\text{rank } A = 1 \quad \underbrace{\begin{bmatrix} 1 & 2 & 0 & 0 \\ 1 & 2 & 0 & 0 \end{bmatrix}}_{} \; 1$$

$Ax = 0$

$$\begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

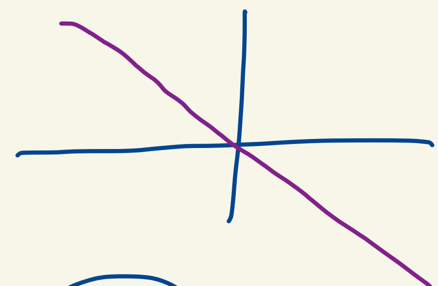
$$x_1 + 2x_2 = 0$$

$$c\begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

rank A + null A = 2

1   1   2

1   3!!   4



$x_1 = 0$, $x_2 = 0$ a solution? Yes!! is $x_1 = x_2 = 0$ the only solution.

$$\text{rank}(AB) \le \min(\text{rank}\,A, \text{rank}\,B)$$

$$AB = A[B_{c_1} \;.\; . \quad B_{c_n}]$$

$$= [AB_{c_1} \quad . \;.\; . \quad AB_{c_n}]$$

$$\mathcal{C}(AB)$$

rank $(AB) \le$ rank A

rank $(AB) \le$ rank B

$AB_{c_1} \ldots AB_{c_n}$: every column of AB is a linear combination of column of A

$\mathcal{C}(AB) \subset \mathcal{C}(A) \Rightarrow \text{rank}(AB) \le \text{rank}\,A.$

every Row of AB a linear comb. of rows of B

$\mathcal{R}(AB) \subset \mathcal{R}(B) \Rightarrow \text{rank}(AB) \le \text{rank}\,B$